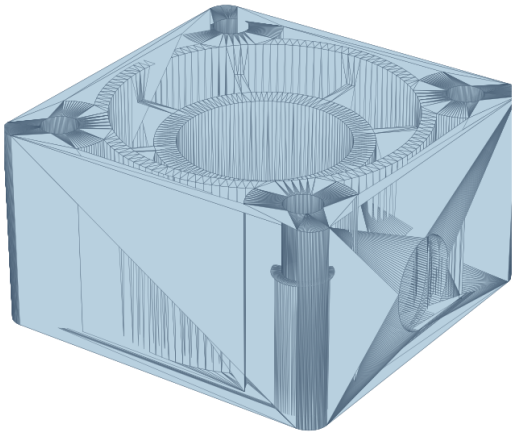
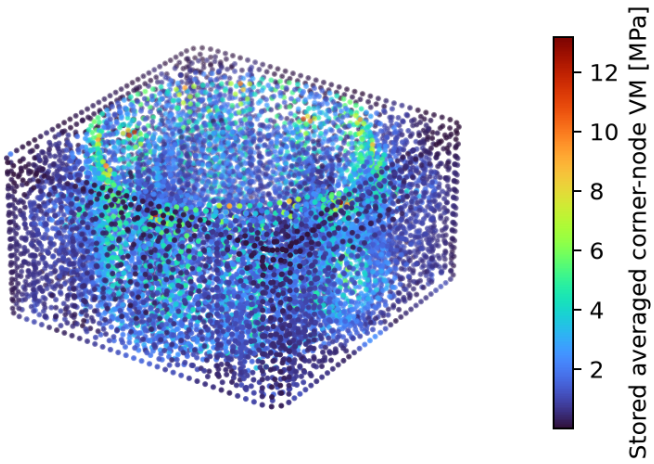
Pneumatic-cylinder cover: original CAD and FEM

13 October 2026. This extension adds a pneumatic-cylinder cover and a titanium wing source with saved FEM data. A simple dogbone source is separated as a buckling/scoring control. Parking/barrier mechanisms add original CAD and drawings but remain held because no numerical FEM reference was located.



Read-only source STEP view: independently valid BREP, one solid and 102 faces, envelope 62 x62 x34.6 mm. Original native Discovery DSCO is retained. This is a moderately detailed pressure-cover part, not a complex moving assembly. Display tessellation is not a new solver mesh.



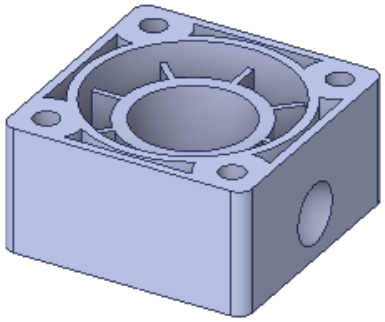
Read-only visualization of the existing ANSYS result: original finite averaged corner-node stresses shown at their stored coordinates. Derived von-Mises peak is 13.185679 MPa; missing stresses are omitted and preserved as NaN/mask. The point plot is an inspection of downloaded fields, not an original publisher screenshot or a new FEM solve.

Original 2022 R1 static RST has 68,720 nodes, 46,105 elements and one saved set. Source pressure is 0.5 MPa. Native CAD, Discovery model, Workbench project, source deck/log and full original RST are included. One part/one pressure case contributes one candidate, irrespective of node count.

Original archive extraction passed all 23 member CRCs and sizes; STEP validity and native field copies were checked. Geometry/load-face mapping, material plausibility and mesh convergence still need review before benchmark admission.

Source: original publisher repository. Original filenames, source tree, Git blob hashes and SHA-256 retained.

Cover: numerical references and source constraints



Unchanged original Discovery thumbnail: extracted from the native source file; source STEP was independently imported. A thumbnail is not a verified dimensioned drawing.

Source FEM reference / setup	Value or finding
Max translation norm	0.009226294 mm
Peak derived averaged von Mises	13.185679 MPa
Finite stress nodes	10,655 of 68,720
Missing stress nodes	58,065; NaNs and mask retained
Pressure / solver units	0.5 MPa; /units,MPA: mm, N, MPa
Material definition	Engineering Plastic; deck E22,000 MPa, nu0.3
Fixed support	Source component _FIXEDSU contains only node 843, all DOFs fixed
Additional restraints	Selected _FRICSUX and _FRICSUZ nodes have UX=0 and UZ=0 respectively

Source deck does not fix every mounting face. Its single-node fixed support and directional frictionless restraints must be matched exactly when recreating boundary conditions. The pressure acts through source surface-element selection; verify its geometric scope rather than assuming the whole exterior is loaded.

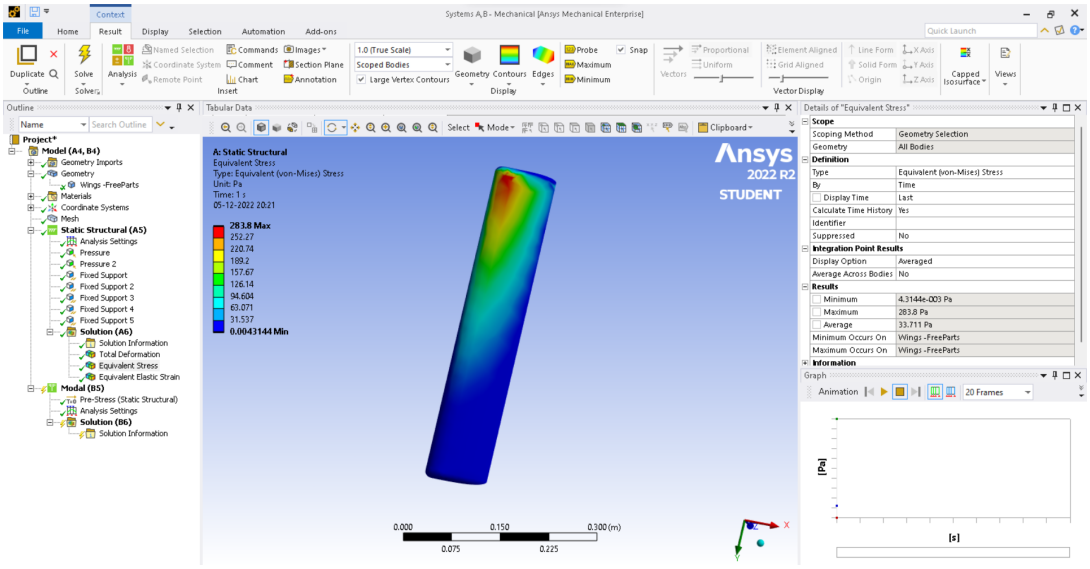
Engineering Plastic is the publisher material name. E 22 GPa is kept exactly as supplied; no corrected material value is substituted. The field values reflect those original assumptions and should not be interpreted as a validated operating pressure or product safety rating.

extracted-original-fields.npz retains original node IDs, mesh coordinates, translations, six-component averaged stresses and missing-value mask. Von Mises is derived from the stored tensor, not a separately solved quantity or raw integration-point stress. Field recovery uses ansys-mapdl-reader 0.56.0; repeated reads agree exactly but do not establish independent solver verification.

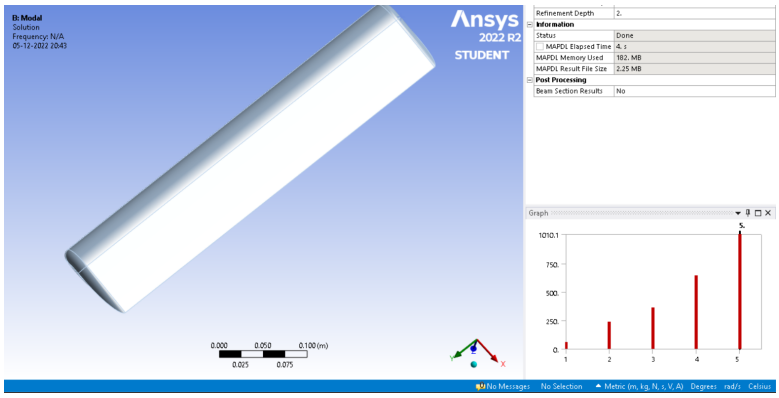
Supplied-CAD-to-FEM candidate only. No verified dimensioned manufacturing drawing is available for this cover. Required review: CAD/CAE revision and transform, selected support/pressure faces, material card interpretation, convergence and target recovery convention.

Source: original publisher repository. Original filenames, source tree, Git blob hashes and SHA-256 retained.

Titanium wing: saved static and modal fields



Original publisher static FEM figure: 283.8 Pa equivalent-stress maximum. Recovered averaged tensor gives 283.802497 Pa, consistent with the displayed rounding. Original native DesignModeler FFF.agdb is included; no neutral STEP/IGES was uploaded and proprietary geometry has not been independently imported.



Original publisher modal/geometry view: five natural frequencies are displayed. Saved RST values are 58.785703, 240.508460, 364.832438, 643.713571 and 1010.110750 Hz. All five normalized mode arrays and exact frequencies are copied to modal-original-fields.npz.

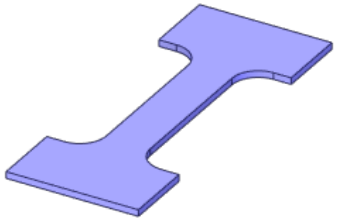
Existing saved solution	Inventory / reference
Static	3,239 nodes /1,863 elements; one set; max U 1.6433948e-8 m; stress finite 565 nodes, missing 2,674
Prestressed modal	Same mesh; five frequencies/mode vectors; arbitrary mode normalization
Original source cards	MKS titanium E 96 GPa, nu0.36, density4620 kg/m3; two source pressure objects with SFE component 3=0.1 Pa; five fixed-support groups

This is a simple standalone airfoil wing, with static and prestressed modal behavior. Very low 0.1 Pa source pressure explains the small displacement/stress scale; it is not an aircraft flight-load reference. Four distributed partition RST files are not four extra cases.

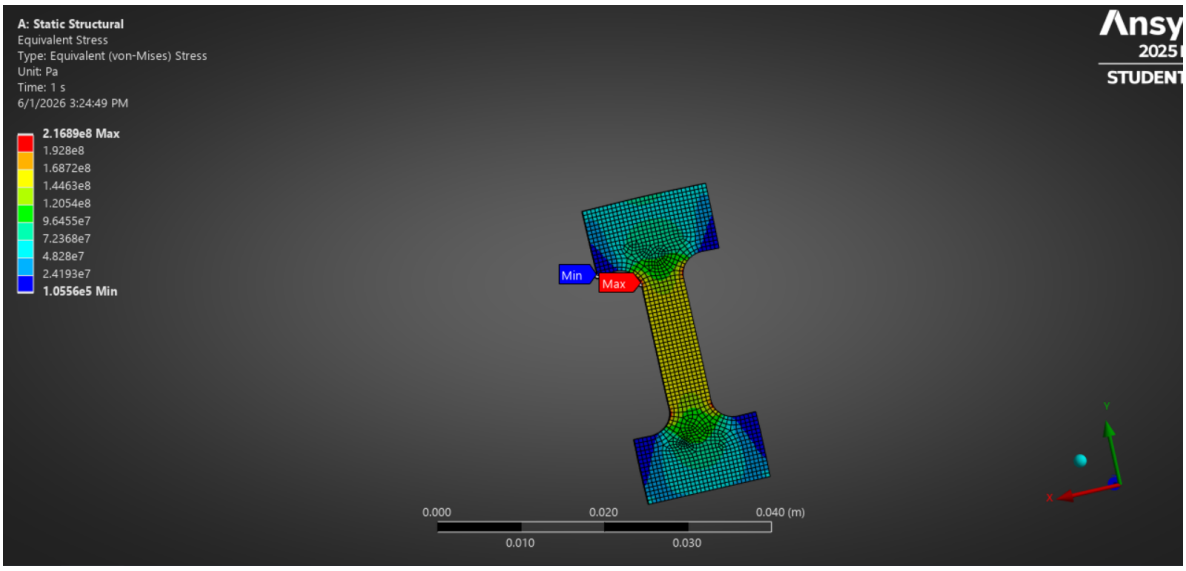
The publisher title mentions flutter, but these saved results do not provide a coupled aerodynamic flutter eigenproblem or critical airspeed. Natural frequencies alone cannot score flutter prediction. Native CAD import, load/support topology and geometry revision remain pending. No new CAD/mesh or solver run.

Source: original publisher repository. Original filenames, source tree, Git blob hashes and SHA-256 retained.

Dogbone: simple buckling control, separate count



Original stored Discovery geometry preview: native SYS.dsco retained. Source describes 38 mm length, 15 mm end width, 5 mm waist, 3 mm fillet radius and3 mm thickness. No compatible independent solid import or locally synthesized STEP is claimed.



Original publisher static stress figure: maximum 216.89 MPa. Companion fatigue plots report minimum life 19,101 cycles and safety factor 0.39743. These are source plot references; fatigue S-N curve, mean-stress/load-ratio convention and exact case mapping need review.

Buckling mode	Original dimensionless multiplier	Source rounded
1	0.7817336297	0.78173
2	2.0094410291	2.0094
3	5.7321480567	5.7321

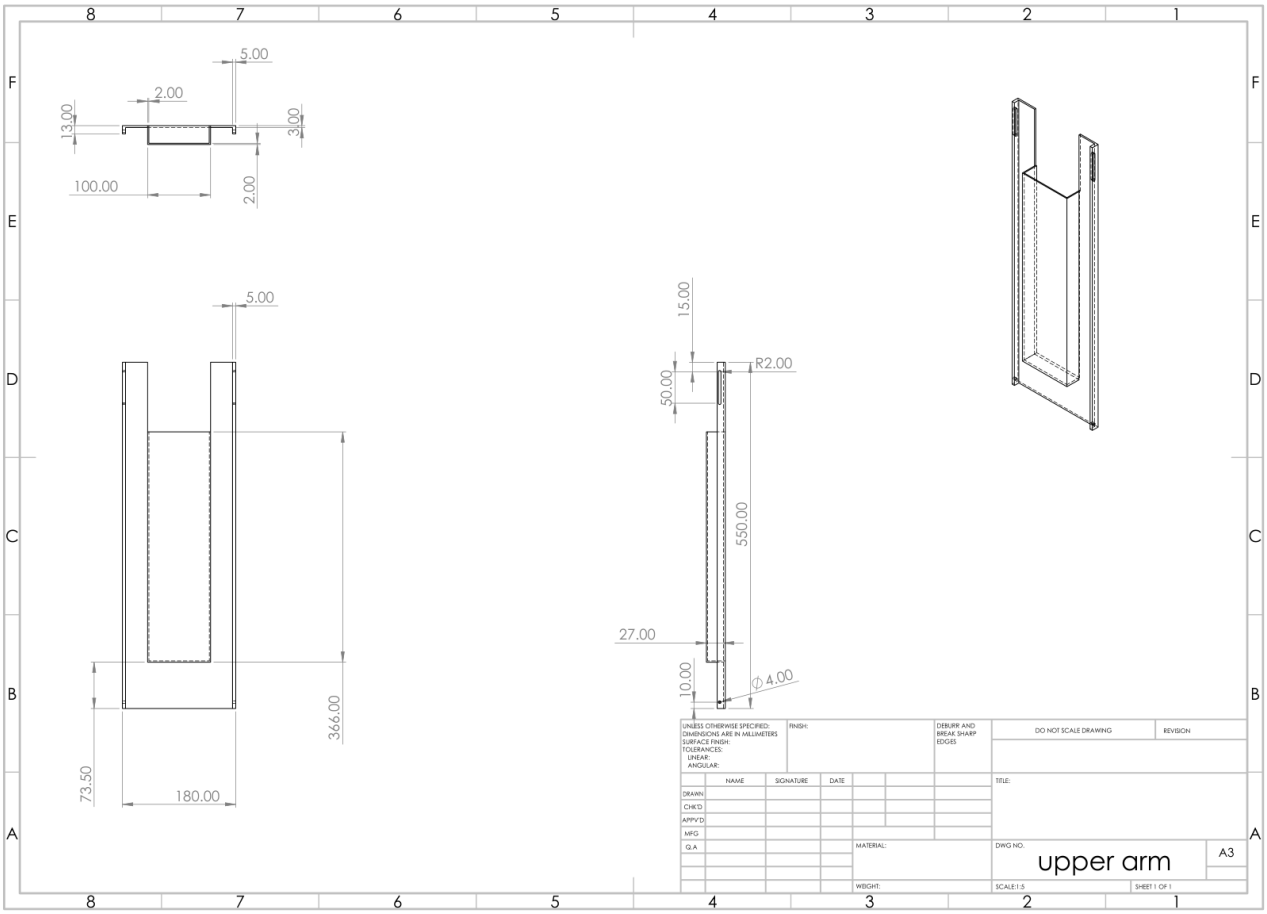
Original static and buckling RST files each describe 17,260 nodes /3,030 elements. Source uses 50 MPa compressive pressure and steel E 200,000 MPa. Header multipliers match README rounded values. These are load factors, not Hz; mode deformation is normalized, not actual displacement.

README claims 1 mm static deformation without a supplied static deformation plot. Mode 3 screenshot amplitude 0.88889 mm does not substantiate that claim. No legacy 2025 R2 full-field extraction is used. Static/buckling files are two linked solutions in one simple geometry; fatigue plots are not a third saved RST.

One calibration/control source; zero complex-task count. A stress peak below nominal yield does not establish buckling safety, since the first source multiplier is below 1. Native import and compatible field validation remain pending.

Source: original publisher repository. Original filenames, source tree, Git blob hashes and SHA-256 retained.

Parking systems: original drawings, FEM held



Original upper-arm engineering drawing: dimensioned orthographic/section/isometric views and title block specifying millimeters. This is one of nine original one-page drawing PDFs for the parking-lock mechanism. Material/tolerance/completeness and matching native parts require review.

Original CAD archive / evidence	Independent inspection / status
Barrier motor/shaft STEP	Valid 25 solids /466 faces; partial mechanism scope
Parking upper-assembly STEP	Valid 15 solids /224 faces; full native assembly dependencies pending
Parking cap STEP	Valid 1 solid /11 faces; envelope 215 x107.574 x580mm
Rotary parking system	Original SolidWorks parts/assemblies; includes third-partyreference/catalogimages
Paired numerical FEM candidates	0 located; no RST/RTH, solver history, numerical report or result plot

Publisher describes ANSYS validation and explicitly omits raw result files. The three downloaded archives contain CAD and reference images/drawings, but no located numerical FEM result. CAD complexity does not fill this gap. These sources remain held; neither three projects nor nine drawings increase paired benchmark counts.

Original native assembly/part files and nine dimensional PDFs are included for future matching to a published FEM source. No mechanism was recreated or solved. Repository MIT notice retained; external reference photos/catalog drawings do not automatically inherit that license.

Source: original publisher repository. Original filenames, source tree, Git blob hashes and SHA-256 retained.

Collection inventory and benchmark preparation

Added evidence	Count / qualification
Pneumatic-cylinder cover	1 valid neutral CAD / raw static-result candidate
Titanium wing	1 native geometry source /2 saved static-modal candidates; proprietary CAD import pending
Dogbone control	1 simple geometry /2 linked raw solutions ;0 complex-task count
Parking/barrier systems	3 CAD-only projects held ;9 original drawing PDFs unpaired
Certified admissions / model scores	0 / none

Verification: 118 original repository files passed pinned Git blob SHA-1, size and SHA-256 checks. Cylinder RAR extraction passed all 23 original member CRC/size checks. Cover field copies and wing static/five-mode arrays match repeat reads. Buckling multipliers match native headers/source rounding. Missing stresses and original units are retained.

Supplied CAD to FEM: cover is independently valid native STEP. Wing and dogbone have original native CAD files but need compatible import. Freeze exact CAD/CAE transforms, pressure/support scopes, material cards, solver/recovery conventions and tolerances before frontier-model evaluation. Nodal arrays and saved modes are not separate tasks.

Drawing to CAD to FEM: parking-lock drawings supply useful original dimensional information, but no paired numerical FEM reference was located. Keep those assets held until an exact original solve is found. Cover/wing/control previews are not manufacturing drawings. No drawing reconstruction was performed.

Other crawled leads: CarCrashNet publisher says release is pending peer review; advertised 15,567 simulations are not downloaded/admitted. Angled-spar battery wing source ZIP has scripts/notebook and no verified native CAD; its 1.776 GB Results ZIP was not downloaded. Curved post-buckling panel input decks and graphite-foam EnSight meshes are not automatically non-mesh CAD/result pairs.

Provenance / licensing: original source files and archive members remain unchanged. Source programs/macros/journals are retained as text and never executed. Cover, wing and control data-license status unresolved; parking MIT notice has external-image limitations. Local research collection only; no public release or license grant implied.

Two engineering CAD/FEM evidence sources, one separately counted simple control and three unpaired CAD projects. Complexity ranges are stated: cover is a moderate detailed part,wing is a simple standalone component,parking assemblies are complex but unpaired. No locally created CAD/problem, new FEM solve, COMSOL dependency or frontier-model call.

Dataset ZIP includes original CAD/drawings, RSTs, decks/logs, numerical copies, catalogs, source hashes and crawl exclusions. Model benchmark scores have not been measured. Certified admission remains zero; published FEM references are candidates requiring exact problem definition and verification.